Telescoping sums

High school · sequences · Lesson and method · 10

Prerequisites. Fractions, finite sums and indices.

Key point · Ideas and method. A telescoping sum is a sum of consecutive differences. Interior terms cancel: $\sum(a_k - a_{k+1}) = a_1 - a_{n+1}$ for k from 1 to n.
Write the first three terms and the last two to expose cancellation and check endpoints. Term-by-term simplification does not remove the need to specify indices.
A useful decomposition is $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$, valid for $k \geq 1$. Verify it by taking a common denominator.

Example 1. $\sum \frac{1}{k(k+1)}$, from $k = 1$ to n, equals $1 - \frac{1}{n+1} = \frac{n}{n+1}$. The sum is below 1 and tends to 1 as n grows.

Activity 1 · Explore. Compute the first five partial sums, then study the gap to 1. Distinguish the finite sum from its limit.